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Decision Trees and Bagging

Decision trees and bagging are both machine learning techniques used for classification and regression tasks.

Decision Trees: A decision tree is a hierarchical model used to classify or predict outcomes based on a set of rules or conditions learned from training data. The tree is constructed by recursively partitioning the data into subsets based on the feature values that best separate the data points according to the target variable. Each internal node of the tree represents a decision based on a feature, and each leaf node represents a class label or a regression value. Decision trees are interpretable, easy to understand, and can handle both categorical and continuous data. However, they are prone to overfitting and can have high variance.

Bagging: Bagging (short for Bootstrap Aggregating) is an ensemble learning method that combines multiple models to improve prediction accuracy and reduce variance. In bagging, multiple decision trees are trained on different subsets of the training data, obtained by resampling the original data with replacement. Each tree in the ensemble is trained independently, and the final prediction is made by aggregating the predictions of all trees, usually by taking the majority vote (for classification) or the average (for regression). Bagging can reduce the variance of the model, improve generalization, and handle complex decision boundaries. However, it may sacrifice some interpretability.

In summary, decision trees are a simple yet powerful machine learning method that can be used for both classification and regression tasks. Bagging is an ensemble method that uses multiple decision trees to improve prediction accuracy and reduce variance. Bagging can be used with any base model, including decision trees, to improve its performance.

The following links provide additional information on decision trees and bagging

[Decision Trees: An Overview](#)

[Bagging and Random Forest Ensemble Algorithms for Machine Learning](#)

[Predictive Modeling and Statistical Learning](#) (from Yuansi Chen at Duke University)

[Bagging and Random Forest in Machine Learning](#)

